13. Matrix exponential

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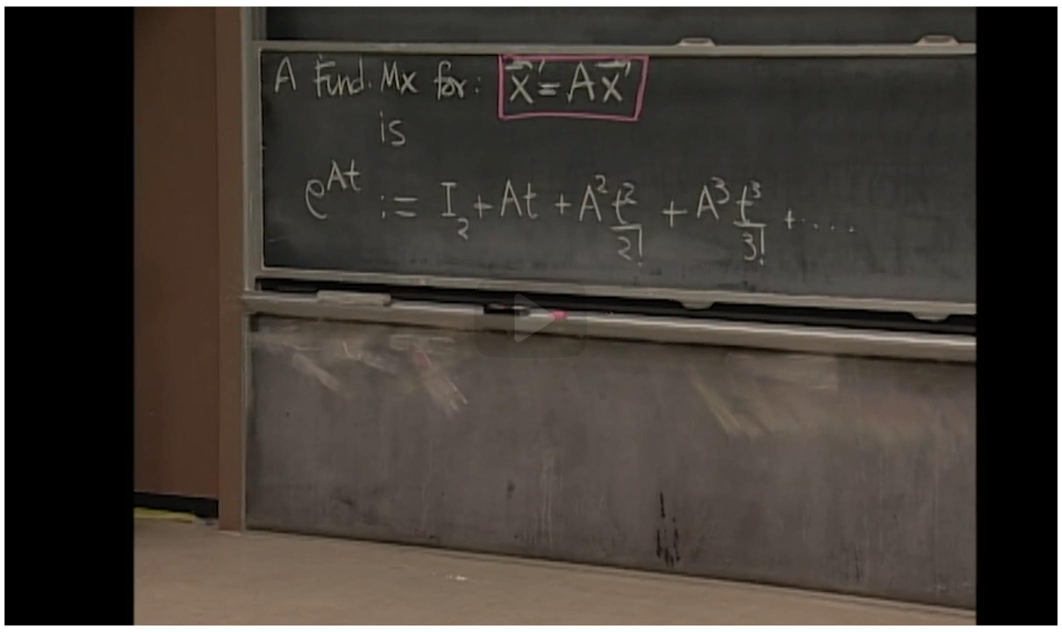
The matrix exponential as a fundamental matrix



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factorial.

What kind of a guy is that?

Well, if A is a 2 by 2 matrix, so is A squared.

How about this?

This is just a scalar which multiplies

every entry of A squared.

And therefore, this is still a 2 by 2 matrix.

That's a 2 by 2 matrix.

This is a 2 by 2 matrix.

No matter how many times you multiply A by itself,

it stays a 2 by 2 matrix.

It gets more and more complicated looking,

but it's always a 2 by 2 matrix.

Now I'm multiplying every entry of that

by the scalar t cubed over 3 factorial.

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We now switch gear to a more theoretical construct, known as the “matrix exponential." The matrix exponential is another way to solve an $n \times n$ linear homogeneous system.

For a real (or complex) number x,

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots.$$

Definition 13.1 For any square matrix $\mathbf{A}$, define the **exponential of the matrix** $\mathbf{A}$ to be the infinite sum

$$e^{\mathbf{A}} := I + \mathbf{A} + \frac{\mathbf{A}^2}{2!} + \frac{\mathbf{A}^3}{3!} + \cdots.$$

So $e^{\mathbf{A}}$ is another matrix of the same size as $\mathbf{A}$.

Theorem 13.2 The function $e^{\mathbf{A}t}$ is a fundamental matrix for the system $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$.

We check the two criteria for fundamental matrices:

- The function $e^{\mathbf{A}t}$ satisfies $\dot{\mathbf{X}} = \mathbf{A}\mathbf{X}$ and
- $\underbrace{\text{its value at } 0}_{I}$ is nonsingular.

Consequence: The general solution to $\dot{\mathbf{x}} = A\mathbf{x}$ is $e^{\mathbf{A}t}\,\mathbf{c}$ where $\mathbf{c}$ is a constant vector.

Compare:

The solution to $\dot{x} = ax$ satisfying the initial condition $x(0) = c$ is $e^{at}c$.

The solution to $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ satisfying the initial condition $\mathbf{x}(0) = \mathbf{c}$ is $e^{\mathbf{A}t}\,\mathbf{c}$.

Question 13.3 If the solution is as simple as $e^{\mathbf{A}t}\,\mathbf{c}$, why did we bother with the method involving eigenvalues and eigenvectors?

Answer

Because computing $e^{\mathbf{A}t}$ is usually hard! In fact, the standard method for computing it (we'll see this later) involves finding the eigenvalues and eigenvectors of $\mathbf{A}$!

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